pamaldi at SemEval-2026 Task 11: Neuro-Symbolic Syllogistic Reasoning via LLM-Guided Structure Extraction and Deterministic Validation

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Abstract

We describe our participation in SemEval-2026 Task 11, Subtask 1: determining the formal validity of syllogisms in English while minimizing the influence of content plausibility. Our system implements a neuro-symbolic pipeline that cleanly separates the neural and symbolic components of syllogistic reasoning. An LLM extracts the formal structure of natural language syllogisms—identifying proposition types (A, E, I, O), the three terms, and the syllogistic figure—while a deterministic symbolic validator checks whether the resulting mood–figure combination is among the 24 classically valid Aristotelian forms. Self-consistency voting across multiple extraction samples and targeted post-processing rules further improve robustness. On the official evaluation, our system achieves 96.34% accuracy, a Total Content Effect (TCE) of 1.02, and a combined score of 56.57. Notably, the near-zero TCE confirms that our system is virtually immune to content bias. Error analysis on the test set reveals that all seven residual errors originate in the neural extraction stage, while the symbolic validator itself introduces no content-dependent failures.

1 Introduction

Large language models (LLMs) have demonstrated remarkable capabilities across a wide range of NLP tasks, yet they exhibit a persistent vulnerability to *content effects* in logical reasoning: they tend to accept logically invalid arguments when the conclusion is plausible, and reject valid arguments when the conclusion seems implausible (Dasgupta et al., 2022; Eisape et al., 2024). This entanglement of formal reasoning with world knowledge is a fundamental limitation that undermines the reliability of LLMs in tasks requiring strict logical inference.

SemEval-2026 Task 11 (Valentino et al., 2026) addresses this challenge by evaluating systems on syllogistic reasoning across four subtasks of increasing complexity, from English-only binary classification to multilingual settings with irrelevant premises. In this paper, we focus on **Subtask 1**: predicting the formal validity of categorical syllogisms in English, where the official ranking metric jointly rewards high accuracy and penalizes content bias through the Total Content Effect (TCE).

Our key insight is that syllogistic validity is a purely structural property: it depends entirely on the *mood* (the combination of proposition types A, E, I, O) and *figure* (the arrangement of terms across premises), not on the truth or plausibility of the premises. This motivates a neuro-symbolic architecture that decomposes the problem into two stages: (1) a *neural extraction* stage, where an LLM parses natural language syllogisms into their formal components, and (2) a *symbolic validation* stage, where a deterministic checker verifies whether the extracted mood–figure pair belongs to the set of 24 classically valid Aristotelian forms.

Our system achieves a combined score of 56.57 on the official evaluation, with 96.34%

accuracy and a TCE of only 1.02—confirming near-complete immunity to content effects. Error analysis reveals that all residual errors originate in the extraction stage, while the symbolic validator itself is inherently content-agnostic.

2 Background

2.1 Task Description

SemEval-2026 Task 11 (Valentino et al., 2026) investigates the ability of computational systems to perform formal logical reasoning independently of content plausibility. Subtask 1 requires predicting the binary validity label of syllogisms presented in English natural language. Each syllogism consists of two premises and a conclusion, and is annotated with both a `validity` label (the target) and a `plausibility` label (reflecting real-world commonsense). The training set contains 200 instances in English, balanced across four categories: valid–plausible, valid–implausible, invalid–plausible, and invalid–implausible.

2.2 Evaluation Metrics

Systems are evaluated by three metrics: (1) **Overall Accuracy** (ACC), the percentage of correct validity predictions; (2) **Total Content Effect** (TCE), a composite score measuring the average accuracy difference due to plausibility across logical–plausibility conditions, where a lower TCE indicates greater robustness to content bias; and (3) the **Primary Ranking Metric**, computed as:

$$\text{Score} = \frac{\text{ACC}}{1 + \ln(1 + \text{TCE})} \tag{1}$$

This metric rewards high accuracy while smoothly penalizing content bias.

2.3 Categorical Syllogisms

A categorical syllogism consists of exactly three terms—the *major term* (P), *minor term* (S), and *middle term* (M)—distributed across two premises and a conclusion. Each proposition takes one of four forms: **A** ("All S are P"), **E** ("No S are P"), **I** ("Some S are P"), or **O** ("Some S are not P"). The *mood* of a syllogism is the ordered triple of proposition types (e.g., AAA, EIO). The *figure* (1–4) is determined by the position of the middle term across the two premises (Copi et al., 2014).

Under the traditional Aristotelian interpretation with *existential import*—where universal propositions presuppose the existence of their subjects—there are exactly 24 valid mood–figure combinations distributed across four figures.

2.4 Related Work

Recent work has documented content effects in LLM reasoning (Dasgupta et al., 2022; Eisape et al., 2024; Bertolazzi et al., 2024), with detailed analyses across syllogistic forms (Ozeki et al., 2024; Seals and Shalin, 2024; Kim et al., 2025). Approaches to mitigate these effects include activation steering (Valentino et al., 2025; Maraia et al., 2026), quasi-symbolic chain-of-thought (Ranaldi et al., 2025; Lyu et al., 2023; Xu et al., 2024), and LLM–symbolic theorem prover combinations (Quan et al., 2024). Our work differs in using a strict separation between neural extraction and deterministic validation, rather than relying on LLM-generated reasoning chains.

3 System Overview

3.1 Design Evolution

Our final system emerged from an iterative exploration of four distinct architectures, each addressing limitations uncovered in the previous one. We re-executed all approaches on the official test set (191 instances) after gold labels were released; results are reported in Table 1.

Prolog Code Generation with Reflexion. Our initial approach prompted the LLM to generate executable SWI-Prolog programs encoding each syllogism's logical structure, with a Reflexion self-correction loop (Shinn et al., 2023) providing up to 3 retry attempts on execution failure. While conceptually elegant, this approach achieved only 90.58% accuracy with a high TCE of 8.49 (combined score: 27.87). Only 110 of 191 test instances succeeded on the first attempt, and the LLM could encode content bias directly into the generated rules.

Neuro-Symbolic with Full LLM Extraction. The key insight was that *validity checking does not require code generation*. We redesigned the pipeline as a two-stage system: the LLM extracts syllogistic structure (types, terms, figure) as a pure NLU task, while a deterministic module checks whether the form belongs to the 24 valid Aristotelian forms. With self-consistency voting ($k = 4$), extraction reflexion, and figure verification, this achieved 97.91% accuracy and TCE of 2.13 (combined score: 45.74). Error analysis revealed two bottlenecks: incorrect figure computation by the LLM and I/O proposition type confusion.

Simplified Extraction with Deterministic Figure. To eliminate figure errors—the most frequent category—we further reduced the LLM's role: it extracts *only* proposition types and terms, while the figure is computed **deterministically** from the middle term's position. With $k = 3$ samples and no reflexion, this achieved 97.38% accuracy and a slightly lower TCE of 2.08 (combined score: 45.81). Despite marginally lower accuracy, the better TCE yields a higher combined score.

Incremental Meta-Learning. We also experimented with automatic prompt refinement: training errors were analyzed every 50 samples and used to generate an enhanced extraction prompt with few-shot examples and guidelines (prompt grew from 15,878 to 24,585 characters across 4 checkpoints). While this helped on training data, it achieved only 95.29% accuracy on the test set with a TCE of 6.25 (combined score: 31.97), suggesting that the meta-learned prompt overfit to training error patterns.

Convergence. The submitted system combines the most effective elements: simplified NLU extraction with deterministic figure computation, self-consistency voting, targeted post-processing rules, and an LLM fallback. The central finding across all approaches is that errors consistently originate in neural extraction, never in deterministic validation, and that more deterministic components yield lower content effects.

Approach	**Acc.**	**TCE**	**Score**
Prolog + Reflexion	90.58	8.49	27.87
Incr. Meta-Learning	95.29	6.25	31.97
Full LLM Extraction	97.91	2.13	45.74
Simpl. Det. Figure	97.38	2.08	45.81
Final (submitted)	**96.34**	**1.02**	**56.57**

Table 1: Test set results (191 instances) across explored approaches. The final submitted system achieves the best combined score due to its lowest TCE. First four rows are post-hoc re-evaluations.

3.2 Final Pipeline

Our submitted system follows a two-stage neuro-symbolic pipeline: (1) an LLM extracts the syllogism's formal structure as three independent samples, aggregated by majority vote; (2) post-processing rules correct common errors (I/O confusion, double negation, compound terms); (3) a deterministic symbolic validator checks membership in the 24 valid Aristotelian forms; (4) when extraction fails entirely, a direct LLM fallback provides a validity judgment.

3.3 Neural Extraction Stage

Given a syllogism in free text, the LLM extracts the **three terms** (S, P, M), the **proposition types** (A, E, I, O) for each statement, and the **figure** (1–4) is computed deterministically from the middle term's position. The prompt includes detailed I/O distinction examples and step-by-step figure computation procedures.

Self-Consistency Voting. We sample $k = 3$ independent extractions at temperatures 0.1, 0.3, and 0.5, selecting the most frequent mood–figure combination. When all three disagree, the first extraction is used.

Figure	Valid Moods
1	AAA, EAE, AII, EIO, AAI, EAO
2	EAE, AEE, EIO, AOO, EAO, AEO
3	IAI, AII, OAO, EIO, AAI, EAO
4	AEE, IAI, EIO, AEO, EAO, AAI

Table 2: The 24 valid syllogistic forms under Aristotelian logic with existential import.

3.4 Symbolic Validation Stage

The deterministic checker verifies whether the extracted mood–figure pair belongs to the 24 valid Aristotelian forms (Table 2). This component contains no learned parameters and makes no reference to semantic content, ensuring complete immunity to content effects by construction.

3.5 Post-Processing and Fallback

Between extraction and validation, rule-based post-processing corrects systematic errors: **I/O disambiguation** detects negation markers to fix I→O misclassifications; **double negation handling** recognizes patterns like "no X are not Y" as type A; **compound term normalization** ensures consistent matching of complex noun phrases. When extraction fails entirely (~5–10% of instances), the system falls back to a direct LLM validity judgment, which achieves lower accuracy than the symbolic path (§5).

4 Experimental Setup

Data and Model. Subtask 1 provides 200 training and 191 test English syllogisms, balanced across four validity–plausibility conditions. We used the training set for prompt development and error analysis without fine-tuning. Our extraction and fallback components use Qwen3-32B[1] with few-shot prompting. For self-consistency, we sample $k = 3$ extractions at temperatures 0.1, 0.3, and 0.5. The pipeline is implemented in Python 3.11 using `boto3` for LLM inference; all other components are pure Python.[2]

[1]Qwen3-32B accessed via Amazon Bedrock API.

[2]Code: `https://github.com/pamaldi/pamaldi_semeval_task11`

	Plausible	Implausible
Valid	95.83% (46/48)	95.83% (46/48)
Invalid	97.87% (46/47)	95.83% (46/48)

Table 3: Accuracy across validity–plausibility subgroups (test set). Near-uniform performance yields TCE = 1.02. Overall: 96.34% accuracy, combined score 56.57.

Configuration	Acc.	TCE
Full system	97.0%	2.78
– self-consistency	94.0%	4.95
– I/O post-processing	96.0%	$\sim$4.0
– 24 → 15 forms (no EI)	93.0%	$\sim$5.5
Baseline (initial prompt)	70.0%	>10

Table 4: Ablation study on the training set (200 instances). "–" indicates removal of a component.

Development. The system was developed through iterative error analysis on the training set. Key iterations included adding I/O disambiguation rules, restoring the full 24 Aristotelian forms (with existential import) after discovering the task adopts the traditional interpretation, and fixing self-consistency voting and fallback parsing bugs.

5 Results

Our system achieves 96.34% accuracy (184/191), a TCE of 1.02, and a combined score of 56.57 on the official evaluation. Table 3 reports accuracy across the four validity–plausibility subgroups: the system achieves near-identical accuracy across all conditions, confirming that the symbolic validation stage successfully decouples validity judgments from content plausibility. Of 191 test instances, 184 were resolved via the symbolic path and 7 via the LLM fallback.

The errors comprise 4 false negatives and 3 false positives, spread across subgroups without systematic bias (2–2–1–2).

5.1 Ablation Study

Table 4 shows the impact of key components on training set performance (200 instances).

The adoption of the full 24 Aristotelian forms (with existential import) and the self-consistency voting mechanism provide the largest individual gains. The I/O post-processing rules yield a modest but consistent improvement.

6 Error Analysis

We analyzed all 7 prediction errors on the test set. All errors are attributable to the neural extraction stage; the symbolic validator introduces no errors on correctly extracted structures.

The errors comprise 4 false negatives (valid syllogisms predicted as invalid, including *Barbara* AAA-1 and *Fresison* EIO-4) and 3 false positives (invalid syllogisms mapped to valid forms due to proposition type misidentification). The dominant error types are: proposition type confusion (I$\leftrightarrow$O, E$\leftrightarrow$O), and figure computation errors from incorrect middle term identification. Detailed error breakdowns are in Appendix C.

Crucially, the 7 errors are distributed nearly uniformly across plausibility conditions (2–2–1–2), confirming that extraction errors are driven by linguistic complexity rather than content plausibility. This yields the near-zero TCE of 1.02.

During development, we also identified 5 annotation inconsistencies in the training set where valid Aristotelian forms (e.g., Celarent EAE-1, Baroco AOO-2) were labeled as invalid. These did not appear in the test set.

7 Discussion

The strict separation of neural extraction from symbolic validation offers three key advantages: (1) the symbolic component is provably immune to content effects; (2) errors are localizable to either extraction or validation; (3) every prediction is interpretable via the extracted mood–figure pair.

A critical design decision was adopting the Aristotelian interpretation with existential import (24 valid forms) rather than modern logic (15 forms). Initial experiments with 15 forms showed substantially lower accuracy ($\sim$93%), confirming that the task adopts the traditional convention.

In principle, a perfect symbolic validator achieves TCE $= 0$. Our TCE of 1.02 is close to this minimum; the small residual arises because extraction errors are not perfectly uniformly distributed across plausibility conditions. Further reducing TCE requires improving the extraction component, perhaps through content-agnostic prompting or symbolic preprocessing.

Our approach is limited to classical categorical logic. The LLM extraction stage remains the primary bottleneck for syllogisms with complex noun phrases, double negations, or non-standard phrasing. The fallback LLM path ($\sim$4% of cases) reintroduces potential content sensitivity. Additionally, our system only addresses Subtask 1 (English binary classification).

8 Conclusion

We presented a neuro-symbolic system for SemEval-2026 Task 11 that achieves 96.34% accuracy with a combined score of 56.57 and a TCE of only 1.02, demonstrating near-complete immunity to content effects. All seven residual errors originate in the neural extraction stage, distributed uniformly across plausibility conditions. The system also identified annotation inconsistencies in the training data, highlighting the interpretability advantages of the neuro-symbolic paradigm. Future work includes extending to multilingual settings (Subtask 3) and handling irrelevant premises (Subtasks 2 and 4).

References

Leonardo Bertolazzi, Albert Gatt, and Raffaella Bernardi. A systematic analysis of large language models as soft reasoners: The case of syllogistic inferences. In *Proceedings of EMNLP*, 2024.

Irving M Copi, Carl Cohen, and Kenneth McMahon. *Introduction to Logic*. Pearson, 14th edition, 2014.

Ishita Dasgupta, Andrew K Lampinen, Stephanie CY Chan, Hannah R Sheahan, Antonia Creswell, Dharshan Kumaran, James L McClelland, and Felix Hill. Language models show human-like content effects on reasoning tasks. *arXiv preprint arXiv:2207.07051*, 2022.

Tiwalayo Eisape, Michael Tessler, Ishita Dasgupta, Fei Sha, Sjoerd van Steenkiste, and Tal Linzen. A systematic comparison of syllogistic reasoning in humans and language models. In *Proceedings of NAACL*, 2024.

Geonhee Kim, Marco Valentino, and Andre Freitas. Reasoning circuits in language models: A mechanistic interpretation of syllogistic inference. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 10074–10095, Vienna, Austria, July 2025. Association for Computational Linguistics.

Qing Lyu, Shreya Havaldar, Adam Stein, Li Zhang, Delip Rao, Eric Wong, Marianna Apidianaki, and Chris Callison-Burch. Faithful chain-of-thought reasoning. In *Proceedings of AACL*, 2023.

Giuseppe Maraia, Marco Valentino, Fabio Massimo Zanzotto, and Leonardo Ranaldi. Abstract activation spaces for content-invariant reasoning in large language models. *arXiv preprint arXiv:2602.02462*, 2026.

Kentaro Ozeki, Risako Ando, Takanobu Morishita, Hirohiko Abe, Koji Mineshima, and Mitsuhiro Okada. Exploring reasoning biases in large language models through syllogism: Insights from the NeuBAROCO dataset. In *Findings of ACL*, 2024.

Xin Quan, Marco Valentino, Louise Dennis, and André Freitas. Verification and refinement of natural language explanations through LLM-symbolic theorem proving. In *Proceedings of EMNLP*, 2024.

Leonardo Ranaldi, Marco Valentino, and André Freitas. Improving chain-of-thought reasoning via quasi-symbolic abstractions. In *Proceedings of ACL*, 2025.

Tyrese Seals and Valerie Shalin. Evaluating the deductive competence of large language models. In *Proceedings of NAACL*, 2024.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. *Advances in Neural Information Processing Systems*, 36, 2023.

Marco Valentino, Geonhee Kim, Dhairya Dalal, Zhixue Zhao, and André Freitas. Mitigating content effects on reasoning in language models through fine-grained activation steering. *arXiv preprint arXiv:2505.12189*, 2025.

Marco Valentino, Leonardo Ranaldi, Giulia Pucci, Federico Ranaldi, and André Freitas. Semeval-2026 task 11: Disentangling content and formal reasoning in language models. In *Proceedings of the 20th International Workshop on Semantic Evaluation (SemEval-2026)*, 2026.

Jundong Xu, Hao Fei, Liangming Pan, Qian Liu, Mong Lee, and Wynne Hsu. Faithful logical reasoning via symbolic chain-of-thought. *arXiv preprint arXiv:2405.18357*, 2024.

A Extraction Prompt Template

The extraction prompt instructs the LLM to:

1. Identify the two premises and the conclusion.

2. For each proposition, determine the type (A, E, I, or O), paying special attention to the distinction between I (“Some S are P”) and O (“Some S are *not* P”).

3. Identify the three terms (S, P, M) based on their distribution: S appears in the minor premise and conclusion; P appears in the major premise and conclusion; M

It.	Change	Acc.	Score
0	Baseline prompt	70%	20
1	Structured extr.	90%	33
2	Self-consistency	94%	33.8
3	I/O post-proc.	96%	40
4	24 Arist. forms	97%	41.7
5	Compound terms	97.5%	45
6	Double negation	98%	48

Table 5: Development progression on the training set (200 instances). Approximate values shown.

appears in both premises but not the conclusion.

4. Compute the figure (1–4) based on the position of M in the premises, following a step-by-step procedure with worked examples for each figure:
    - Figure 1: M is subject in P1, predicate in P2
    - Figure 2: M is predicate in both premises
    - Figure 3: M is subject in both premises
    - Figure 4: M is predicate in P1, subject in P2
5. Output the mood (e.g., "EIO") and figure (e.g., "4") in a structured format.

B Complete List of Development Iterations

Table 5 summarizes the progression of system performance through iterative development on the training set.

C Test Set Error Details

Table 6 lists all 7 errors on the official test set with abbreviated syllogism content.

#	Type	Sub.	Likely Cause
1	FN	V+I	EIO-4 (Fresison): figure error
2	FN	V+P	AAA-1 (Barbara): extraction fail
3	FN	V+I	EAO-1 (Celaront): type error
4	FP	I+I	AEA-1 $\rightarrow$ valid: type error
5	FN	V+P	AEO-4: figure error
6	FP	I+I	AOE-2 $\rightarrow$ AOO-2: O/E confusion
7	FP	I+P	AEA-1 $\rightarrow$ valid: type error

Table 6: All 7 test set errors. FN=false negative (valid$\rightarrow$invalid), FP=false positive (invalid$\rightarrow$valid). V=Valid, I=Invalid, P=Plausible.